

EGB322: Engine Performance Analysis Report

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1. Management Summary

A performance experiment and thermodynamic analysis were conducted on the Perkins 404D-22 2.2L diesel engine under various operating conditions. The engine produced a peak power output of 22.9 kW and a peak torque of 118 N.m. Testing revealed the engine achieved a peak thermal efficiency of 29.7% and a peak mechanical efficiency of 90.1%. The average specific fuel consumption across test conditions was 287.3 g/kWh.

2. Introduction

2.1 Intended use of this document

This report details the results and data recorded from an experiment conducted on a four-cylinder (4-stroke) 404D-22 Perkins 2.2L diesel engine. The experiment was designed to measure and understand the engine's performance and thermodynamic behaviour across various operating conditions. Data recorded from the experiment was translated into key operating parameters including BMEP, SFC, A-F Ratio, volumetric efficiency, and thermal efficiency, and compared against industry benchmarks and published literature.

An energy balance was performed to quantify energy losses within the engine, and in-cylinder pressure and volume data were analysed to determine the indicated power and mechanical efficiency. Additionally, an ideal air-standard diesel cycle was fitted to the P-V data, using theoretical assumptions, to describe the real power cycle and calculate key parameters such as cut-off ratio and heat input per cylinder, supporting the engine performance analysis.

2.2 Background

Internal combustion engines (ICE's) have served society as integral power generation devices for nearly 150 years and have been studied extensively, uniting chemical, electrical and mechanical engineering together. These devices have been the backbone of transport, agriculture, mining, construction, power generation and industrial equipment. One family of ICE's is the diesel engine. One of the most notable people in the development of these devices was Rudolf Diesel. The German engineer, in 1892, proposed a new type of engine which relied on fuel being injected into high temperature and high-pressure gas for combustion. His invention allowed doubling the efficiency of other engines available at the time.

The 4-stroke diesel cycle consists of 4 key stages. During the intake stroke, the piston and connecting rod move down, drawing in fresh air into the cylinder. Next, in the compression stroke, the air is compressed and about 20 crank angle degrees before top dead centre (TDC), fuel is injected into the high pressure, high temperature gas and is instantly combusted. The high-pressure explosion drives the piston down in what is known as the power stroke. Just before the piston reaches bottom dead centre (BDC) the exhaust valves open and high pressures cause a rapid blowdown of the burnt gases. After BDC the piston moves back to TDC, exhausting the remaining gases out of the

cylinder. At TDC the process repeats, drawing in fresh air into the cylinder [1]. The following schematic shows the 4-stroke cycle:

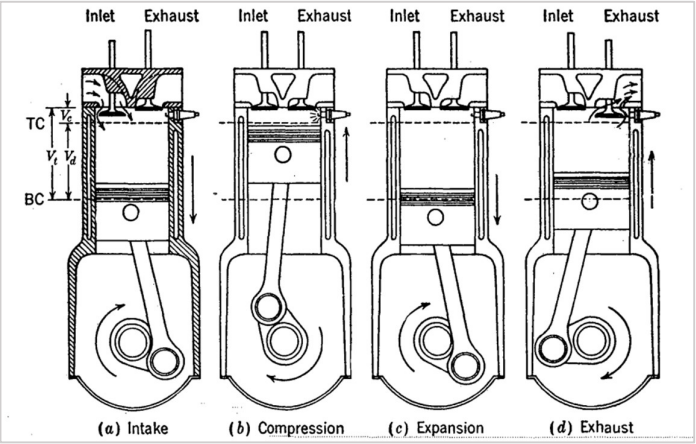


Figure 1. The four-stroke cycle (source: [2])

From the ideal Carnot cycle, it is evident that the ideal efficiency of a heat engine is dependent on the temperature ratio between the hot and cold reservoirs. Below is the carnot efficiency:

$$\eta_{Carnot} = 1 - \frac{T_L}{T_H}$$

The diesel cycle was designed to optimise thermal efficiency and low-end torque. By injecting fuel after compressing the air, the engine can achieve higher compression ratios, leading to higher pressures and temperatures before combustion. This results in a higher temperature ratio, contributing to higher thermal efficiency, though it remains well below the carnot efficiency due to real world irreversibilities.

To analyse the Diesel cycle, real processes are often approximated using ideal thermodynamic assumptions to simplify calculations. In the ideal Diesel cycle, the working fluid is pure air and is treated as an ideal gas and the system is modelled as a closed system. The compression stroke is said to compress adiabatically, with no heat transfer to the surroundings and with no friction losses (Isentropic compression). During the power stroke, combustion (heat addition) is modelled as a constant pressure process since diesel fuel burns gradually throughout the stroke. After combustion, the gas expands adiabatically and reversibly performing work on the piston. At BDC, heat rejection occurs at constant volume, unlike the real exhaust stroke which expels burnt gases in an open system. These idealizations allow simplified analysis of the cycle.

3. Test Engine and Instrumentation

3.1 Engine Specifications

The engine used in the experiments was a 2.2L, 4-cylinder, naturally aspirated industrial diesel engine. The main specifications are given in Table 1.

Table 1. Engine specifications for the 404D-22 Perkins engine.

404D-22 Perkins 2.2L Diesel Engine	
Bore	84 mm
Stroke	100 mm
Displacement	0.002217 m^3
Compression Ratio	23.3:1
Cooling system	Liquid
Combustion system	Indirect Injection

3.2 Instrumentation

The following instrumentation was used in the engine performance experiment.

Table 2. List of instruments and equipment used in the experiment.

Instrument	Purpose/Use
Dynamometer and electric motor	Measure engine power output
U-Tube Manometer	Intake air flow measurement.
Cooling water flow meter	Measure flow rate of cooling water.
Thermocouples	Measure coolant and exhaust gas temperatures.
In-Cylinder P-V data logger	Record pressure and volume changes in the cylinder.
DynoLog Software	Display and control all relevant engine parameters.

3.3 Experiment Setup and Test Rig

The engine was tested in a well-ventilated room at 21°C using a custom-built engine performance test rig. The test rig consists of a metal frame mounted on wheels, with the test engine, dynamometer, and electric motor aligned in series. The engines operating conditions were set using a control panel mounted on a separate frame. The control panel also reported engine load, rpm, power, throttle position and ambient temperature. A separate laptop was connected to the thermocouples TC1-TC4 and was used to record the cooling water and exhaust gas temperatures. K-type thermocouples were used, with a normal temperature range of approximately -200 °C to 1260 °C and a limit of error around ±2.2°C [3]. Figure 2 below shows the test rig setup.

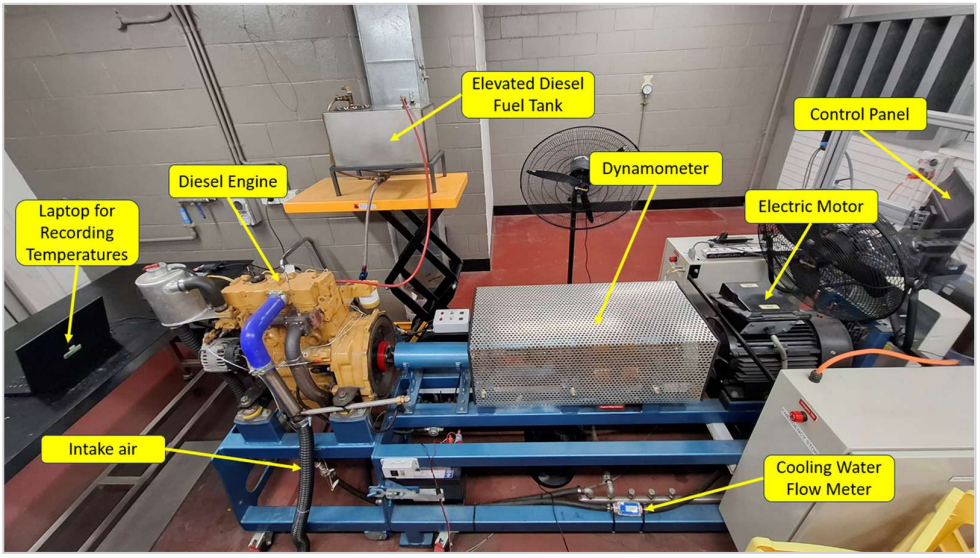


Figure 2. Engine test rig setup with ancillary equipment. Note: air supply tank is not shown.

Air was supplied to the engine without forced induction form an air tank venting to the atmosphere. The pressure difference between the air tank and the atmosphere was measured using a U-tube manometer. The pressure difference was used to calculate the air flow into the engine. Cooling water was circulated through the engine block and a heat exchanger by an engine mounted water pump. A separate cooling water circuit was pumped through the heat exchanger, to remove heat, where inlet and outlet temperatures were recorded.

3.4 Method

The engine was turned on and allowed to heat up to operating temperature. The RPM and throttle dials were set on the control panel to the initial test conditions: 1400 rpm and 100% load. The engine was given approximately 2-3 minutes to reach thermal equilibrium. The measurements for this test condition were then recorded and tabulated. The previous steps were then repeated for the other test conditions. The engine was then allowed to cool down unloaded before it was turned off.

3.5 Control Volume Schematic

For the thermodynamic analysis of the engine, the combustion chamber and exhaust pipe were selected as the control volume for study. Figure 3 below shows the control volume schematic with the relevant energies crossing the system boundary (shown as a dashed line).

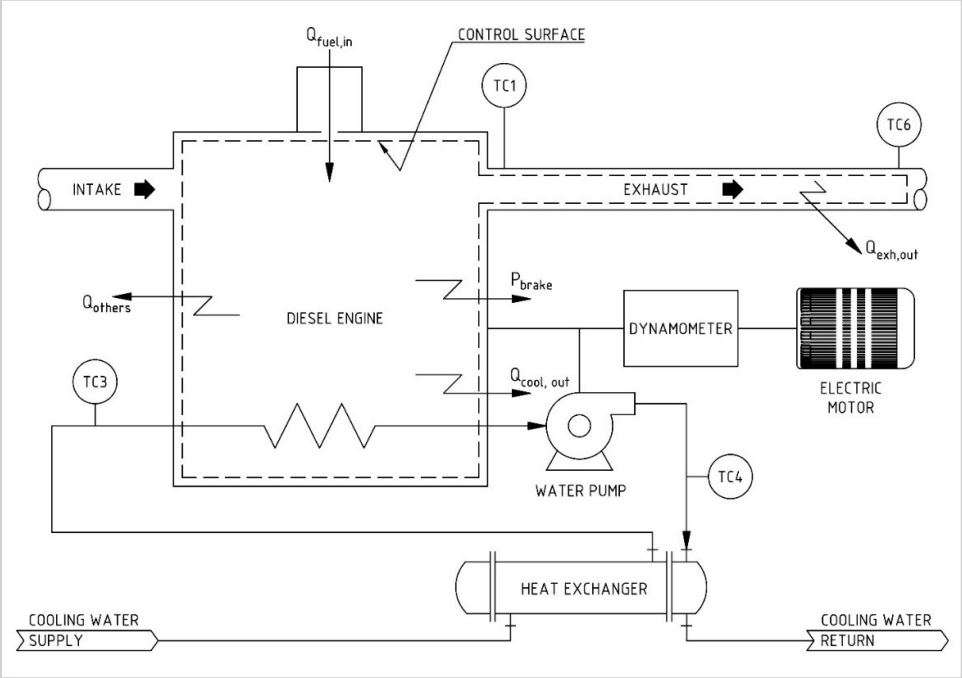


Figure 3. Control volume schematic showing energies crossing the system boundary. Dashed line shows the control surface selected for analysis.

4. Results (Data and Analysis)

Five key operating parameters were calculated to understand the engines performance under varying rpm and load. The experiments were carried out in three different rpm ranges. Under each of these rpm cases the engine was tested under 50% load and 100% load. The results of the engine performance test are presented in this section. Data are presented in tables and figures to highlight key trends. To analyse the performance of the engine, the combustion process was assumed to be steady-flow and the time between loading cases is assumed to be sufficient for the engine to reach thermal equilibrium.

4.1 Experiment Data (Measurements)

The following table presents the raw data recorded from the engine experiments.

Table 3. Table of recorded measurements from the engine experiment.

Engine parameters	Values						Units
Nominal speed	1400	1400	1800	1800	2000	2000	rpm
Engine load	50	100	50	100	50	100	%
Measured speed	1390	1390	1793	1792	2000	2000	rpm
Throttle position	68.0	100.0	85.0	100	96.0	100	%
Break Power	8.70	17.7	10.6	22.1	11.6	22.9	kW
Torque	59.0	122	55.0	118	56.0	109	N.m
Temperatures							

TC3 - Water in temp.	21.90	21.26	22.90	22.30	24.10	23.40	°C
TC4 - Water out temp.	26.90	31.93	31.10	34.90	33.20	37.50	°C
TC5 - Cal. In temp.	174.0	302.0	202.0	360.0	214.0	367.0	°C
TC6 - Cal. Out temp	117.0	143.0	127.0	218.0	140.0	230.0	°C
TC1 - Exh. In temp	242.6	471.5	271.6	508.3	289.4	512.7	°C
TC2 - Exh. Out temp	218.6	408.0	251.5	451.6	267.5	461.8	°C
Flow rates							
Cooling water flow rate	51.5	40.4	41.0	40.4	40.4	40.4	L/min
Fuel supply rate	0.0540	0.0980	0.0590	0.118	0.0620	0.124	L/min
Air							
Manometer el. difference	122	112	202	194	255	230	mm

4.2 Method of calculation

This section provides the calculation methods for the five key operating parameters used in evaluating the engine's performance. This section also outlines the energy balance calculations and the procedure used to analyse the PV-data.

4.2.1 Brake Mean Effective Pressure (BMEP)

$$\mathrm{BMEP} = \frac{2\pi \cdot T \cdot n_r}{V_d}$$

Where:

- n_r = Number of revolutions per cycle
- T = Brake torque
- V_d = Total engine displacement volume

Example calculation (1400 rpm, 100% load):

$$\mathrm{BMEP} = \frac{2\pi \cdot T \cdot \mathrm{n}}{V_d} = \frac{2\pi \cdot 122\ \mathrm{N.m} \cdot 2\ \mathrm{ul}}{0.002217\ m^3} \cdot \left(\frac{1\ MPa}{10^6\ Pa}\right) = 0.692\ (6.92\ Bar)$$

4.2.2 Specific Fuel Consumption (SFC)

$$SFC = \frac{\dot{m}_{fuel}}{P_{brake}}$$

Where:

- $\dot{m}_{fuel}$ = Mass flow rate of the diesel fuel
- P_{brake} = Brake power output of the engine

Example calculation (1400 rpm, 100% load):

$$\dot{m}_{fuel} = \dot{V} \cdot \rho = 0.0980\ L/min\ \cdot 860\ kg/m^3 \cdot \left(\frac{1\ m^3}{1000\ L}\right) \cdot \left(\frac{1\ min}{60\ s}\right) = 1.405 \times 10^{-3}\ kg/s$$

$$SFC\ = \frac{\dot{m}_{fuel}}{P_{brake}} = \frac{1.405 \times 10^{-3}\ kg/s}{17.7\ kW} \cdot \left(\frac{1000\ g}{1\ kg}\right) \cdot \left(\frac{3600\ kJ}{1\ kWh}\right) = 285.7\ g/kWh$$

4.2.3 Mass flow rate of air supplied to engine

The mass flow rate of the air supplied to the engine is calculated using the recorded elevation difference in the manometer fluid level, the orifice diameter, and orifice flow equation. Figure 4 below shows the air tank and U-tube manometer setup.

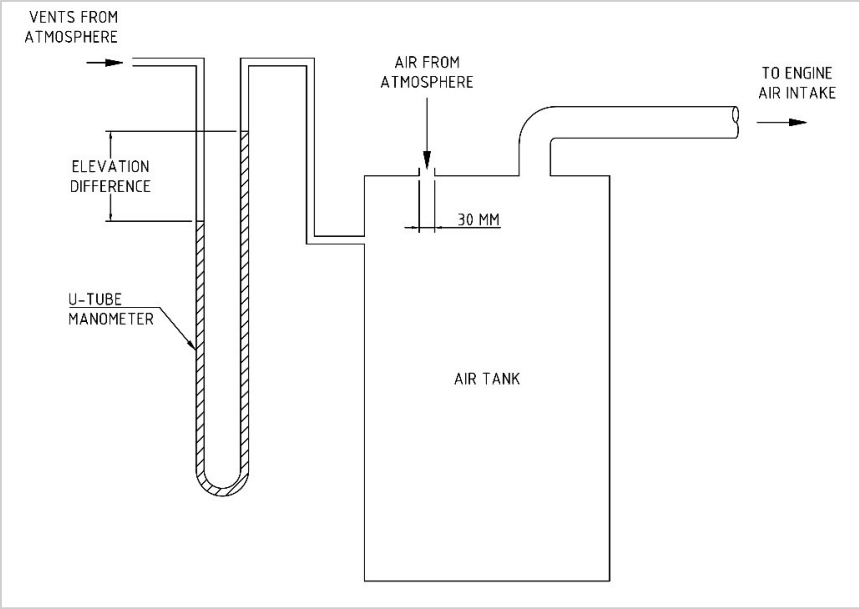


Figure 4. Engine air flow measurement setup.

First the pressure difference is calculated from the recorded elevation difference:

$$P_2 - P_1 = (\rho_{man} - \rho_{air}) \cdot g \cdot (h_2 - h_1)$$

Where:

- ρ_{man} = Density of the fluid in the manometer tube
- $h_2 - h_1$ = Recorded elevation difference of the manometer fluids

Following the pressure difference calculation, the volume flow rate of the air is calculated using the orifice flow equation derived from Bernoulli's equation:

$$\dot{V}_{air,in} = C_o A_0 \sqrt{2 \cdot \frac{(P_2 - P_1)}{\rho_{air}}}$$

Where:

- C_o = Orifice correction factor
- A_0 = Orifice area, (where the orifice diameter is 0.03m)

Example calculation (1400 rpm, 100% load):

$$P_2 - P_1 = (1000\ kg/m^3 - 1.201\ kg/m^3) \cdot 9.81\ ms^{-2} \cdot (112\ mm) \cdot \left(\frac{1\ m}{1000\ mm}\right) = 1097.4\ Pa$$

$$\dot{V}_{air,in} = C_o A_0 \sqrt{2 \cdot \frac{(P_2 - P_1)}{\rho_{air}}} = 0.6 \cdot \frac{\pi \cdot (0.03\ m)^2}{4} \cdot \sqrt{\frac{2 \cdot 1097.4\ Pa}{1.201\ kg/m^3}} = 0.0181\ m^3/s$$

The volume flow rate is then used to calculate the mass flow rate of the air flowing into the engine:

$$\dot{m}_{air,in} = \dot{V}_{air,in} \cdot \rho_{air} = 0.0181\ m^3/s \cdot 1.201\ kg/m^3 = 0.0217\ kg/s$$

4.2.4 Air-Fuel Ratio

$$Air - Fuel\ Ratio = \frac{\dot{m}_{air,in}}{\dot{m}_{fuel}}$$

Where:

- $\dot{m}_{fuel}$ = mass flow rate of diesel fuel
- $\dot{m}_{air,in}$ = mass flow rate of air supplied to the engine

Example calculation (1400 rpm, 100% load):

$$Air - Fuel\ Ratio = \frac{\dot{m}_{air,in}}{\dot{m}_{fuel}} = \frac{0.0217\ kg/s}{1.405 \times 10^{-3}\ kg/s} = 15.4\ ul$$

4.2.5 Volumetric Efficiency

$$\eta_V = \frac{n_r \cdot \dot{m}_{air,in} \cdot 60}{\rho_{air} \cdot V_d \cdot N}$$

Where:

- n_r= Number of revolutions per cycle.
- V_d = Total displacement volume of the engine (m^3)
- N = Actual revolutions per minute (rpm)

Example calculation (1400 rpm, 100% load):

$$\eta_V = \frac{n_r \cdot \dot{m}_{air,in} \cdot 60}{\rho_{air} \cdot V_d \cdot N} = \frac{2 \cdot 0.0217\ kg/s \cdot 60}{1.201\ kg/m^3 \cdot 0.002217\ m^3 \cdot 1390\ rpm} = 0.7036\ ul\ (70.36\ \%)$$

4.2.6 Thermal Efficiency

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_{in}} = \frac{P_{brake}}{\dot{m}_{fuel} \cdot LHV_{fuel}}$$

Where:

- LHV = Lower heating value or energy content per unit mass of diesel fuel (44 MJ/kg)

Example calculation (1400 rpm, 100% load):

$$\eta_{th} = \frac{17.7\ kW}{1.405 \times 10^{-3}\ kg/s \cdot 44\ MJ/kg} \cdot \left(\frac{1000\ W}{1\ kW}\right) \cdot \left(\frac{1\ MJ}{10^6\ J}\right) = 0.286\ ul\ (28.6\ \%)$$

4.2.7 Energy Balance

The energy balance considered the following energies: energy flow into the engine from fuel, energy flow out of the engine via cooling water heat removal, exhaust flow energy and break power output. The remaining unaccounted losses are grouped into a term denoted 'Q_{others}'. The energy balance was calculated using the data from the 1400 and 2000 rpm cases at 100% load only. The energy balance is not calculated at 50% load due to the order of the experiment setup, in that the 50% loading is applied to the engine after the 100% load case, this means that the engine is still cooling down from the hot state it was in at 100% loading and will lead to inaccurate cooling water temperature readings. The steps below outline the methods used for calculating the cooling water heat removal and exhaust flow energy:

$$\dot{E}_{in} - \dot{E}_{out} = \frac{\Delta E_{system}}{dt}$$

Using steady flow assumptions, the change in the system's energy is zero with respect to time, therefore:

$$\dot{E}_{in} - \dot{E}_{out} = \dot{Q}_{fuel,in} - (P_{brake} + \dot{Q}_{cool,out} + \dot{Q}_{ex\ ,out} + \dot{Q}_{other}\ ) = 0$$

Solving for $\dot{Q}_{others}$ allows for calculating the unaccounted losses and thus performing an energy balance.

$$\dot{Q}_{othe} \quad = \dot{Q}_{fuel,in} - (P_{brake} + \dot{Q}_{cool,out} + \dot{Q}_{exh,out})$$

Where:

$$\dot{Q}_{fuel,in} = \dot{m}_{fuel} \cdot LHV_{fuel}$$

$$\dot{Q}_{cool,out} = \dot{m}_{water} \cdot C_{p,water} \cdot (TC4 - TC3)$$

$$\dot{Q}_{exh,out} = (\dot{m}_{air,in} + \dot{m}_{fuel}) \cdot C_{p,ex} \ \cdot (TC1 - TC6)$$

Example calculation (1400 rpm, 100% load):

The specific heat capacity of water at 25° C, along with recorded inlet and outlet temperatures, was used to calculate the energy removed by the cooling water. The mass flow rate of cooling water was calculated from the measured volume flow rate and the density of water at 25° C, which approximates the average temperature between the cooling water inlet and outlet.

$$\dot{m}_{water} = \dot{V}_{water} \cdot \rho = 40.40\ L/min \cdot 997\ kg/m^3 \cdot \left(\frac{1\ m^3}{1000\ L}\right) \cdot \left(\frac{1\ min}{60\ s}\right) = 0.671\ kg/s$$

$$\dot{Q}_{cool,out} = 0.671\ kg/s \cdot 4.18\ kJ/kg.K \cdot (31.93 - 21.26)K = 29.93\ kW$$

The heat removed through the exhaust is then calculated by approximating the exhaust air heat capacity as that of air. The specific heat capacity is determined by averaging the values at the higher and lower temperatures. The Cp values were obtained using Python and the CoolProp library.

$$C_{p,exh} \approx C_{p,air} = \frac{C_{p\,@\,TC1} + C_{p\,@\,TC6}}{2} = \frac{(1.08568 + 1.01618)kJ/kg.K}{2} = 1.05093\ kJ/kg.K$$

It is then calculated as:

$$\dot{Q}_{exh,out} = (0.0217 + 1.405 \times 10^{-3})kg/s \cdot 1.05093\ kJ/kg.K \cdot (471.5 - 143)K = 7.97\ kW$$

Finally, the energy supplied by the fuel is calculated as:

$$\dot{Q}_{fuel,in} = 1.405 \times 10^{-3}\ kg/s \cdot 44\ MJ/kg \cdot \left(\frac{1000\ kJ}{1\ MJ}\right) = 61.82\ kW$$

Applying the energy balance for this load case:

$$\dot{Q}_{other} = 61.82 - (17.7 + 29.93 + 7.97)kW = 6.22\ kW$$

4.2.8 Pressure-Volume Data Analysis

Python, NumPy, Pandas and CoolProp libraries were used to analyse the recorded in-cylinder pressure-volume data. The pressure and volume data is imported from each excel column into two Pandas data frames, and then converted into two NumPy arrays, with one array for volume and one for pressure.

The indicated work per cylinder is calculated from the PV-data by approximating the integral over the cycle using NumPy and the trapezoid() method. The following formula is then used to calculate the indicated power output of the engine.

$$P_{indicated} = \frac{W_{cyl,indicated} \cdot RPM \cdot n_{cyl}}{n_r \cdot 60} = \frac{0.42411\ kJ \cdot 1390\ rpm\ \cdot 4}{2 \cdot 60} = 19.65\ kW$$

Then the mechanical efficiency is calculated using the following formula:

$$\eta_{mechanical} = \frac{P_{brake}}{P_{indicated}} = \frac{17.7\ \text{kW}}{19.65\ kW} = 0.9\ (90\ \%)$$

The indicated mean effective pressure is calculated using the following formula:

$$\text{IMEP} = \frac{\text{W}_{\text{cyl}} \cdot \text{n}_{\text{cyl}}}{\text{V}_{\text{d}}} = \frac{0.42411\ \text{kJ} \cdot 4}{0.002217\ \text{m}^3} \cdot \left(\frac{1\ \text{MJ}}{1000\ \text{kJ}}\right) = 0.765\ \text{MPa}\ (7.65\ \text{Bar})$$

To fit the ideal diesel cycle, the 4 key points are first identified in the real experimental data. These points are (V1, P1), (V2, P2), (V3, P3) and (V4, P4). The points are obtained easily exploiting the fact that the data is ordered in thermodynamic process 0 → 1 → 2 → 3 → 4 → 0. This means that the NumPy array index of points which occur later in the cycle will be greater. V4 corresponds to the maximum volume in the cycle, when the piston is at BDC and hence will be the largest value stored in the NumPy array, and can be easily obtained using the max() method. Based on this maximum value, the element index (position) of this value is returned from within the array. The method returns two indexes, as the value of V4 occurs twice in the array (V4 = V1). Then the values for P4 and V4 are set using the greater value of the two indexes returned since V4 occurs later in the array.

To obtain point 2, the two arrays were trimmed to only include points above P4, to filter the intake and exhaust stoke out. This was done to filter out noise at the beginning of the 1400 rpm data set which showed that the volume decreased slightly before increasing, which may indicate that the experimental data did not start at perfect TDC. V2 is then obtained using the min() method, and the index of that value is returned from the full-size array. The values of V2 and P2 are assigned to variables using the larger index, since point 2 occurs later in the data set. V1 is then easily obtained by the fact that V1 = V4, and the values can be set by using the smaller index returned when searching the data set for V1/V4.

The next step was to find V3, in which the data was trimmed to only include data after the maximum pressure, up until some proportion of P2. The proportion was selected as 0.75. The trimmed data set is then searched for the closest value below and above P2. Once these values are obtained, V3 is determined by interpolating between the lower and upper values to find what volume corresponds with P3.

The four key points are now known, and the polytropic coefficient of expansion and compression is found by solving the following equations:

$$P_1 \cdot {V_1}^{n_{comp}} = P_2 \cdot {V_2}^{n_{comp}}\ and\ \ P_3 \cdot {V_3}^{n_{comp}} = P_4 \cdot {V_4}^{n_{comp}}$$

$$\frac{P_1}{P_2} = \left(\frac{V_2}{V_1}\right)^n \therefore n_{comp} = \frac{\ln(P_1/P_2)}{\ln(V_2/V_1)}\ and\ n_{exp} = \frac{\ln(P_3/P_4)}{\ln(V_4/V_3)}$$

Example calculation (1400 rpm, 100% load):

$$n_{comp} = \frac{\ln(96.92/4726.24)}{\ln(2.89 \times 10^{-5}/0.0005)} = 1.36\ ul$$

$$n_{exp} = \frac{\ln(4726.24/208.96)}{\ln(0.0005/(5.59 \times 10^{-5}))} = 1.42\ ul$$

Once the polytropic coefficient of compression and expansion have been found using the real points, arbitrary points are selected along the compression and expansion strokes to recalculate P1, P2/P3, P4 and V3 for the ideal cycle using the polytropic relationships. The example below selects an arbitrary point along the compression stroke to obtain P2 (ideal), knowing n_{comp} and V2. The second calculation uses P3 (ideal) and an arbitrary point along the expansion stroke to find V3 (ideal).

Example calculation (1400 rpm, 100% load):

$$P_{2,ideal} = \frac{P_{arb} \cdot {V_{arb}}^{n_{comp}}}{{V_2}^{n_{comp}}} = \frac{192.56\ kPa \cdot (0.00029908 m^3)^{1.36}}{(2.89 \times 10^{-5} m^3)^{1.36}} = 4621.8\ kPa = P_{3,ideal}$$

$$V_{3,ideal} = \left(\frac{P_{arb,exp}}{P_{3,ideal}}\right)^{(1/n_{exp})} \cdot V_{arb,exp} = \left(\frac{603.36\ kPa}{4621.8\ kPa}\right)^{(1/1.42)} \cdot 0.00032719\ m^3 = 7.8 \times 10^{-5}\ m^3$$

The remaining unknowns are P4 (ideal) and P1 (ideal) which are found using the polytropic equations:

$$P_{1,ideal} = P_{2,ideal} \cdot \left(\frac{V_2}{V_1}\right)^{n_{comp}} = 4621.8\ kPa \cdot \left(\frac{2.89 \times 10^{-5} m^3}{0.0005\ m^3}\right)^{1.36} = 95.7\ kPa$$

$$P_{4,ideal} = P_{3,ideal} \cdot \left(\frac{V_3}{V_4}\right)^{n_{exp}} = 4621.8\ kPa \cdot \left(\frac{7.8 \times 10^{-5}\ m^3}{0.0005\ m^3}\right)^{1.42} = 330.4\ kPa$$

The cutoff ratio is then calculated as:

$$\frac{V_{3,ideal}}{V_2} = \frac{7.8 \times 10^{-5}\ m^3}{2.89 \times 10^{-5} m^3} = 2.7\ ul$$

The heat input per cylinder is calculated using the constant pressure assumption during the heat addition process, based on temperatures T2 and T3, which are calculated using the ideal gas law state equations ($PV = mRT$) at point 2 and 3. The specific heat capacity is determined using the CoolProp library, which accounts for the high pressure and temperature, and the heat capacities are then averaged across T2 and T3. The mass of the air in the cylinder is determined by the measured air mass flow rate and the engine rpm.

Example calculation (1400 rpm, 100% load):

$$m_{air,cyl} = \frac{\dot{m}_{air} \cdot n_r \cdot 60}{RPM \cdot n_{cyl}} = \frac{0.0217\ kg/s \cdot 2 \cdot 60}{1390 \cdot 4} = 0.000468\ kg$$

$$T_2 = \frac{P_2 \cdot V_2}{m_{air,cyl} \cdot R} = \frac{4621.8\ kPa \cdot 2.89 \times 10^{-5} m^3}{0.000468\ kg \cdot 0.28705\ kJ/kg.K} = 994.3\ K$$

$$T_3 = \frac{P_3 \cdot V_3}{m_{air,cyl} \cdot R} = \frac{4621.8\ kPa \cdot 7.8 \times 10^{-5}\ m^3}{0.000468\ kg \cdot 0.28705\ kJ/kg.K} = 2683.5\ K$$

$$C_{p,avg} = \frac{1.1448 + 1.2837}{2} = 1.2143\ kJ/kg.K$$

$$\dot{Q}_{in,cyl} = \dot{m}_{air,exh} \cdot C_{p,avg}(T3 - T2)$$

$$\dot{Q}_{in,cyl} = (0.0217 + 1.405 \times 10^{-3})kg/s \cdot 1.2143\ kJ/kg.K \cdot (2683.5 - 994.3)K = 11.83\ kW$$

4.3 Processed Results

The following section presents the calculated performance parameters of the test engine along with an energy balance, based on the experimental data in Section 4.1 and the methods outlined in Section 4.2. The key engine operating parameters have been calculated at 1400 rpm through 2000 rpm with 2 loading cases: 50% and 100%. The energy balance only considers 100% load at 1400 and 2000 rpm.

Table 4. Table of calculated engine performance parameters, energy balance and P-V analysis results.

Engine parameters	Values						Units
Nominal speed	1400	1400	1800	1800	2000	2000	rpm
Engine load	50	100	50	100	50	100	%
Output Torque (measured)	59.0	122	55.0	118	56.0	109	N.m
Brake Power (measured)	8.70	17.7	10.6	22.1	11.6	22.9	kW
Key Performance Parameters							
BMEP	0.334	0.692	0.312	0.669	0.317	0.618	MPa
SFC	320.3	285.7	287.2	275.5	275.8	279.4	g/KWh
Air-Fuel Ratio	29.36	15.50	34.57	16.94	36.97	17.55	ul
Volumetric Efficiency	73.71	70.62	73.53	72.10	74.06	70.34	%
Thermal Efficiency	25.55	28.64	28.49	29.70	29.67	29.28	%
Energy Balance							
Energy in fuel		61.82				78.20	kW
Energy to power		17.7				22.9	kW
Energy to cooling water		29.930				39.566	kW
Energy to exhaust		7.970				9.909	kW
Unaccounted losses		6.220				5.828	kW
P-V Analysis							
Indicated Power		19.65				26.86	kW
Mechanical Efficiency		90.1				85.3	%

IMEP	0.765	0.727	MPa
Polytropic Coeff. Expan.	1.423	1.319	ul
Polytropic Coeff. Comp.	1.364	1.406	ul
Cut-off Ratio	2.69	1.78	ul
Available Heat per Cylinder	15.45	19.55	kW
Heat Input per Cylinder	11.87	13.55	kW
Heat Input as Percentage	76.84	69.32	%

4.4 Graphical Results

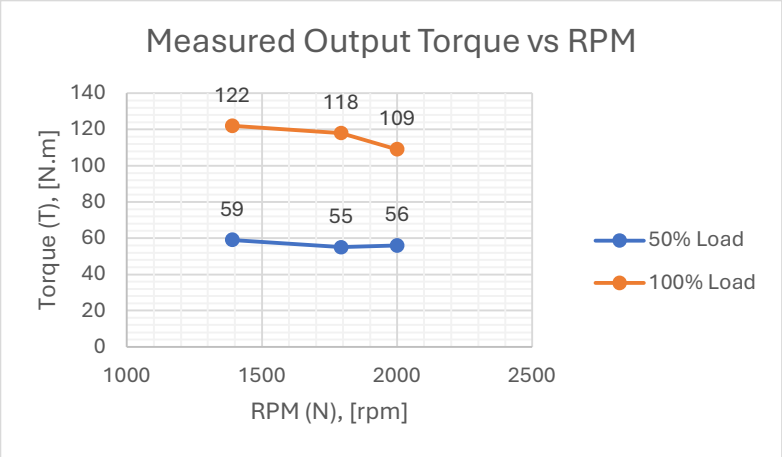


Figure 5. Measured output torque vs engine speed at two loading conditions.

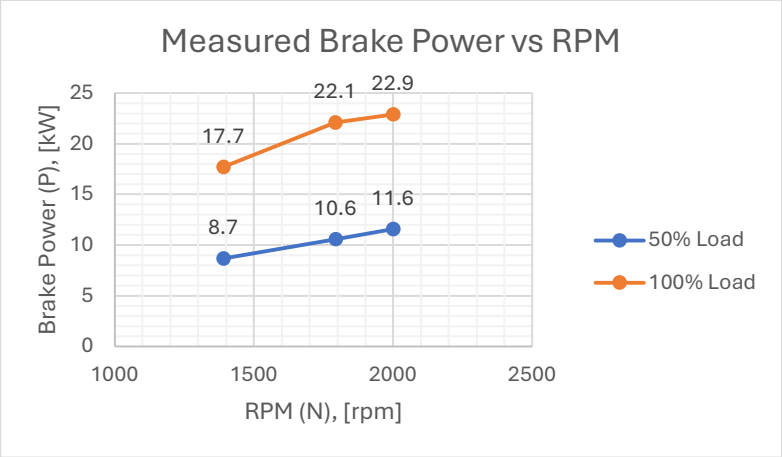


Figure 6. Brake power vs engine speed at two loading conditions.

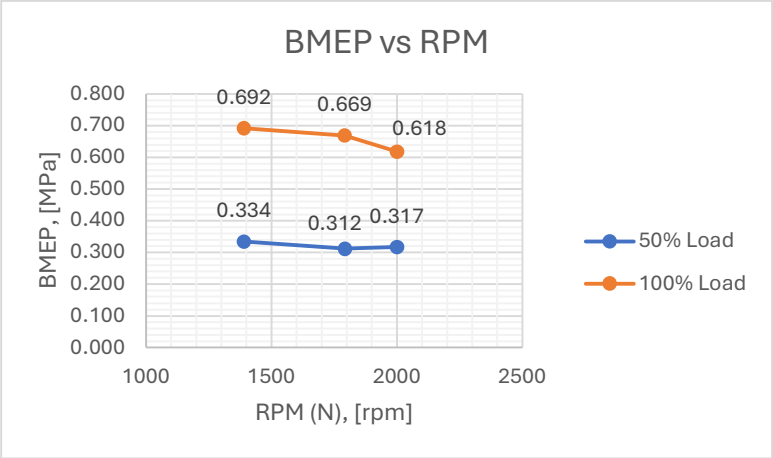


Figure 7. Break mean effective pressure vs engine speed at two loading conditions.

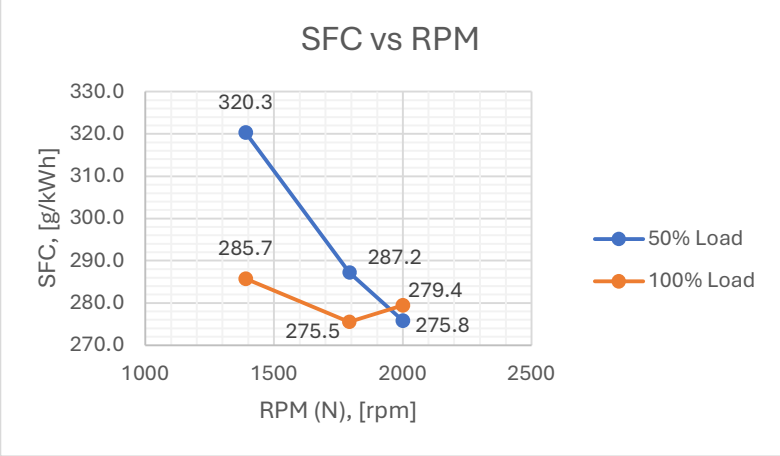


Figure 8. Specific fuel consumption vs engine speed at two loading conditions.

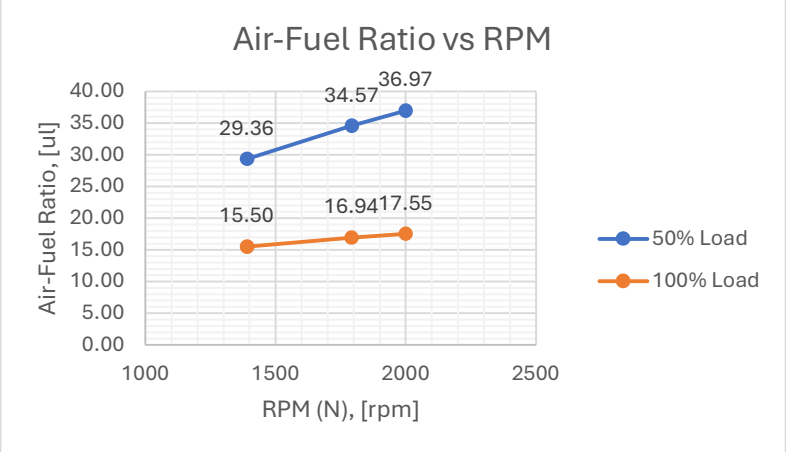


Figure 9. Air to fuel ratio vs engine speed at two loading conditions.

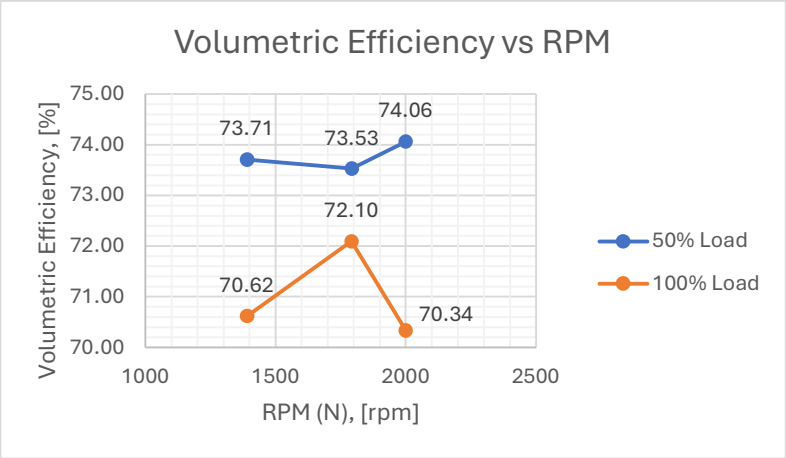


Figure 10. Volumetric efficiency vs engine speed at two loading conditions.

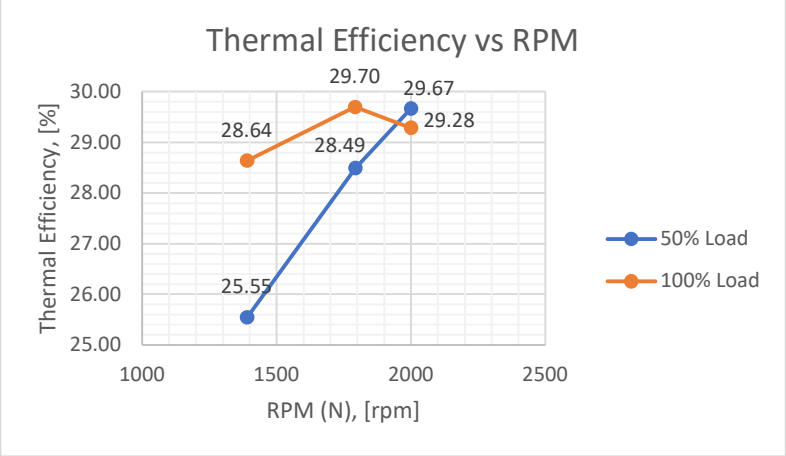


Figure 11. Thermal efficiency vs engine speed at two loading conditions.

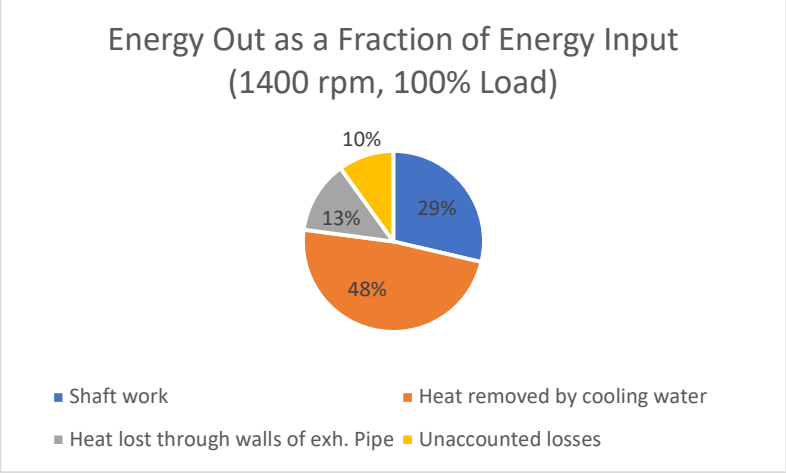


Figure 12. Pie chart showing energy balance of engine at 1400 rpm.

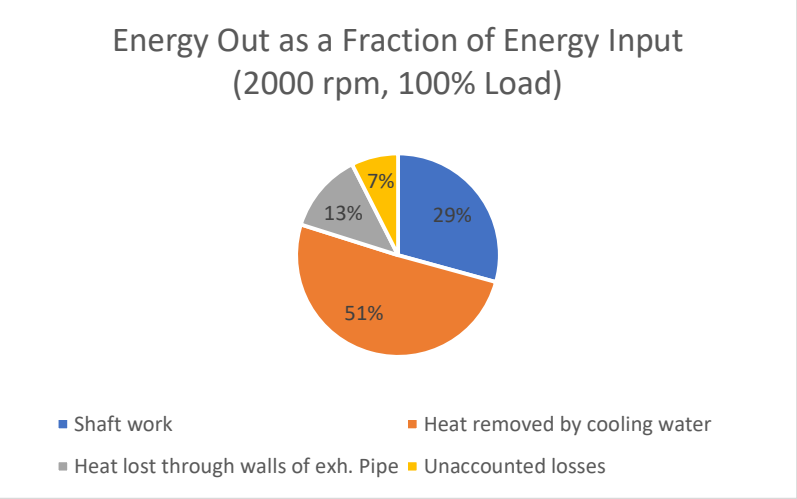


Figure 13. Pie chart showing energy balance at 2000 rpm.

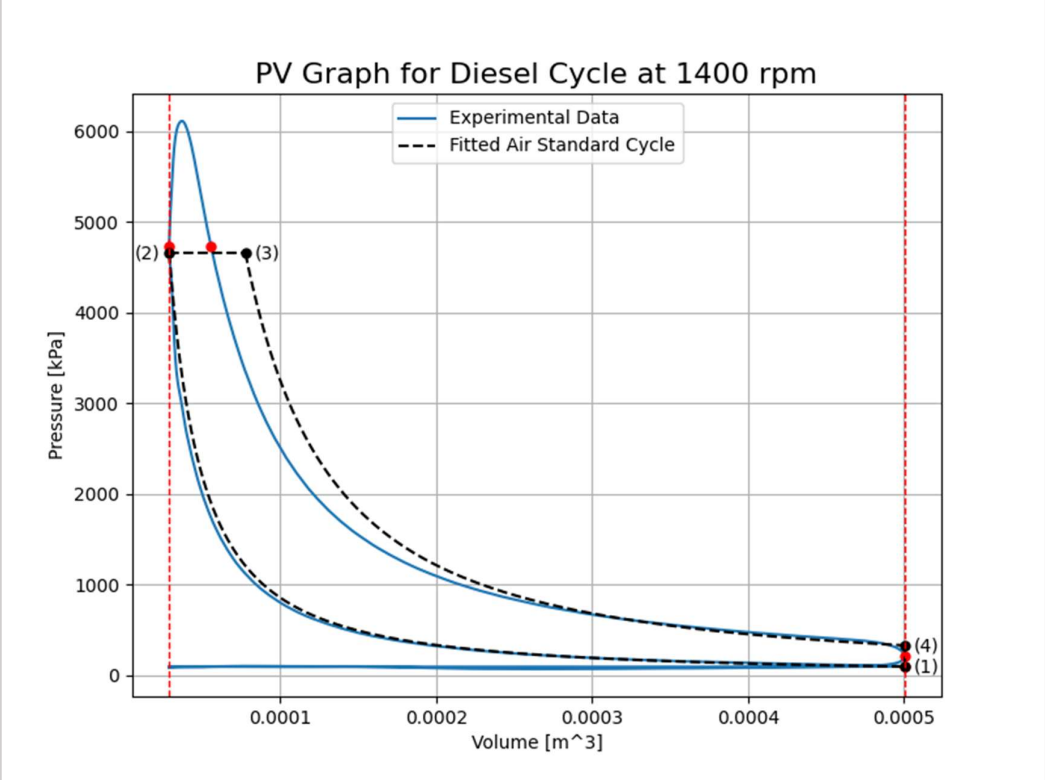


Figure 14. Pressure-Volume Plot for Diesel Cycle at 1400 rpm.

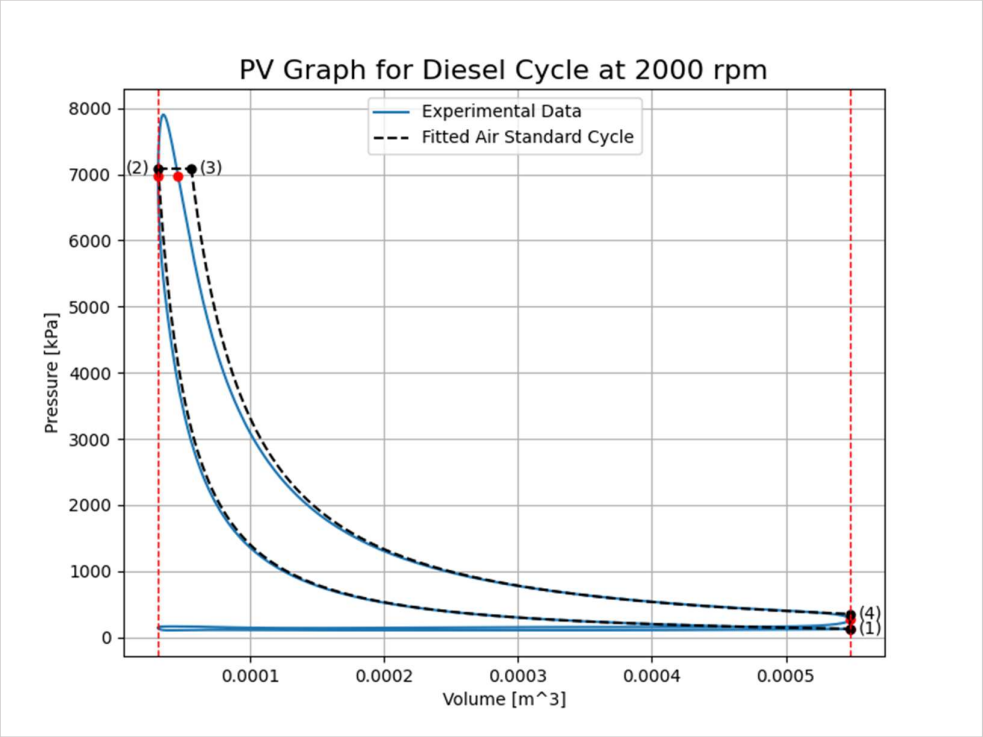


Figure 15. Pressure-Volume Plot for Diesel Cycle at 2000 rpm.

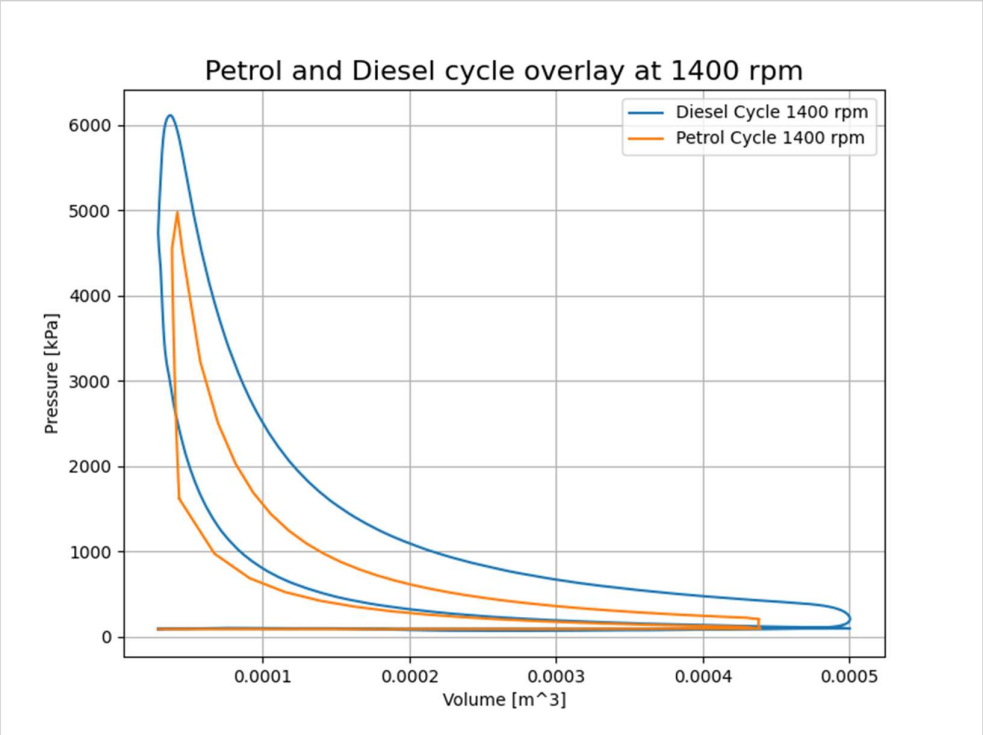


Figure 16. Petrol and Diesel Pressure-Volume Plot Overlay at 1400 rpm.

5. Discussion

5.1 Discussion of Results

In Figure 5 the measured output torque tends to decrease with rpm under full load. The output torque is dependent on the engines ability to effectively convert chemical fuel to mechanical energy. The torque is expected to increase from lower rpm up until a peak torque output, due to better valve timing and more air flow. After the peak, factors such as incomplete combustion, valve timing and reduced volumetric efficiency cause the torque to taper off with a further increase in rpm [1]. The manufacturer's data shows a decrease in torque output after approximately 1800 rpm, which is confirmed by the data recorded from the experiment. The output torque for the 50% load case remains approximately unchanged with varying rpm.

Figure 6 shows a steady increase of brake power with an increase in rpm which matches the manufacturer data. The increase in power output is expected since power is directly proportional to rpm ($P = T \cdot \omega$) and the increase in rpm is much greater than the decrease in torque.

The calculated BMEP for the test engine is shown in Figure 7 and tends to decrease with an increase in rpm under the 100% load case. This aligns with the decrease in engine torque since BMEP is a measure of the average pressure applied to the piston, and the force applied to the piston is proportional to the pressure. In the 50% load case, the BMEP shows a very similar trend to torque in Figure 5. The BMEP recorded at 100% load ranged from 0.62 to 0.69 MPa, which is reasonably close to published values, with typical naturally aspirated diesel engines operating within a range of 0.7 to 1.0 MPa [1].

The SFC illustrated in Figure 8 shows an inverse relationship to thermal efficiency presented in Figure 11 for both load cases. As fuel consumption decreases, thermal efficiency increases. From the formula used to calculate thermal efficiency (see Section 4.2.6), a lower fuel mass flow rate results in higher efficiency for the same power output. The engine requires fuel to produce mechanical power; if the engine produces the same amount of power with less fuel, it operates more efficiently. The measured SFC values are comparable to published values, where naturally aspirated truck diesel engines achieve, at best, a specific fuel consumption of around 210 g/kWh [1].

For both load cases the Air to Fuel ratio steadily increases with engine speed, as shown in Figure 9. This indicates that the mixture becomes leaner as engine speed rises, meaning a greater proportion of air is supplied relative to fuel. The measured air-fuel ratios ranged from approximately 15.5 to 17.6 under 100% load and 29.4 to 37.0 under 50% load. According to Heywood [1], typical air-fuel ratios for diesel engines range from 18:1 at full load to over 40:1 at light load. The results from this experiment fall within the expected range, showing a richer mixture at full load due to a higher fuel flow rate.

The volumetric efficiency of the engine at 100% load increases from 70.6% to 72.1% from 1400 to 1800 rpm, likely due to improved valve timing and increased engine speed. Beyond 1800 rpm, volumetric efficiency decreases to 70.3%. At lower engine speeds the intake process approaches a quasi-steady state, allowing sufficient time for air

to fill the cylinder. At higher speeds, the piston descends faster than air can flow in, which reduces the overall mass of air draw into the cylinder. Under the 50% load case, the volumetric efficiency remains mostly unchanged, ranging from 73.7% to 74.1%. These values are lower than those typically reported for naturally aspirated diesel engines which can reach volumetric efficiencies of about 90% [1].

At 1800 rpm and 100% load, the engine operated at a peak thermal efficiency of 29.7%. The thermal efficiency increases from 1400 to 1800 rpm, due to improved combustion conditions, reduced fuel consumption and increased brake power output. Beyond 1800 rpm, the specific fuel consumption rises, and efficiency declines.

For the 50% load case, the thermal efficiency improves from 25.6% at 1400 rpm to 28.5% at 1800 rpm and further increases to 29.7% at 2000 rpm. Insufficient data was recorded to observe a decrease in thermal efficiency at 50% load. The calculated results from this experiment align reasonably well with values reported in published literature. According to Çengel and Boles [2], the typical brake thermal efficiency of small four-cylinder engines ranges between 35-40%. The measured peak efficiency of 29.7% in this experiment falls slightly below this range, which may be caused by experimental losses and inaccuracies.

Table 4 presents the results of the energy balance, and it is evident that the majority of heat loss occurs through the cooling water, accounting for 48-51% of the heat energy from the diesel fuel in both cases. The next greatest loss was power delivered to the output shaft, followed by heat lost through the exhaust system. Figure 12 and Figure 13 show the distribution of energy from the diesel fuel, with the unaccounted losses reaching 10% of the total energy input in the 1400 rpm case. The unaccounted heat losses are grouped into the term Q_{others}. This term includes all the heat transfer not explicitly measured or calculated in the main balance.

Q_{others} is made up of energy supplied to accessory engine components, such as the energy required to turn the water pump, alternator and fuel pump as well as friction losses. These components are all driven by the engine, and the measured break power is the indicated power minus the friction and accessory losses. Other contributions to this term include radiation and convection to the engine block, heat carried away by the lubrication system, blow by gas energy and other minor losses.

The recorded brake power from the dynamometer is lower for both the 1400 and 2000 rpm cases when compared to the indicated power calculated from the P-V analysis. This is expected due to the fact that the values obtained from the pressure and volume data do not take mechanical losses into account.

Figure 14 and Figure 15 show the experimental Diesel cycle along with the fitted air standard cycle. Higher pressures were recorded in the diesel cycle at 2000 rpm compared to 1400 rpm (6.1 MPa and 8 MPa). The first notable difference between the real and ideal cycle is evident during the 2 → 3 process. The experimentally recorded data shows a continued rise in pressure after point 2, leading to a higher cycle pressure compared to the ideal cycle. This pressure increase occurs due to the injection and subsequent combustion of fuel in the cylinder. In comparison, the air standard cycle assumes a constant pressure heat addition process, since fuel is considered

to be injected gradually and to continue burning as the piston moves downward during the power stroke, causing the increasing cylinder volume to offset the pressure rise. In the real cycle, pressure spikes and irreversibilities occur due to non-uniform combustions, early ignition of premixed fuel, and heat transfer losses.

Another notable difference in Figure 14 and Figure 15 is the process from point 4 to 1. In the real cycle, the heat rejection process occurs as exhaust is expelled out of the cylinder from BDC to TDC. The ideal cycle, assumes a closed system, where heat rejection occurs at BDC as a constant volume process.

Figure 16 shows the petrol cycle compared to a diesel cycle. The petrol cycle starts at a lower volume than the diesel cycle, and both compress to approximately the same volume, indicating that the petrol cycle has a lower compression ratio. The petrol cycle has a peak pressure of around 50 Bar while the diesel cycle reaches higher pressure at approximately 61 Bar. This higher pressure will allow the diesel cycle to achieve higher thermal efficiency. The petrol cycle shows a more abrupt rise in pressure after combustion indicating a near constant volume heat addition, while the diesel cycle shows a pressure increase as volume increases. This represents the constant pressure heat addition.

During the expansion phase both the diesel and petrol cycle show similar curves, the main difference being that the diesel cycle is above the petrol, indicating higher pressure during expansion which encloses a larger loop area. This translates to a higher net work output and higher efficiency in the diesel cycle.

5.2 Experimental Accuracy and Errors

The method used to measure the air flow rate into the engine is subject to potential inaccuracies. The orifice flow equation uses a correction factor which may not account for the hole geometry perfectly. There is a potential for vacuum leaks in the air supply tank and the connecting hoses.

The fuel flow rates in the experiment were not measured values; these were static values provided at the start of the experiment. Since the energy flow into the engine is calculated from the fuel flow rate, and these values are static, the energy flow into the engine does not adapt to the changes in engine rpm. This can lead to inaccuracies in the engine balance. The time required for the engine to reach true equilibrium in between loading cases was unknown, therefore it's possible that the time in between tests was not sufficient. This may lead to variations in the readings.

To reduce errors in future experiments it is recommended to record live fuel consumption with a weight scale and timer. For more accurate air flow measurement it is recommended to remove the need for an orifice correction factor by using a smooth transition opening or a straight pipe before the nozzle to ensure developed flow upstream. The cooling water temperatures and flow rate should be measured from the fluid circulating through the engine block.

6. Conclusion

This report investigated the performance of the Perkins 404D-22 diesel engine under varying operating conditions. The results showed that thermal and volumetric efficiency tend to increase with an increase in rpm, up until a limit, where combustion parameters become less ideal and the engine reduces its performance. The analysis indicates that the engine is likely designed to be operated at around 1800 rpm under full load and at higher rpm under lower loads.

7. References

[1] J. B. Heywood, Internal Combustion Engine Fundamentals, New York, USA: McGraw-Hill, 2018.

[2] Y. A. Cengel and M. A. Boles, Thermodynamics: An Engineering Approach, 10th ed., New York, USA: McGraw-Hill Education, 2020.

[3] A. R. Rogowski, Elements of Internal Combustion Engines, New York, USA: McGraw-Hill, 1953.

8. Appendix

The figure on the right shows the points selected within the real P-V data (in red) and the ideal re-calculated points in black. Note that the real point 1 is difficult to see as it is shadowed by the ideal point 1. Compression begins at BDC which is why point 1 (real) is selected at the end of the intake stroke from 0→1. The heat rejection process in an ideal cycle begins at BDC which is why point 4 is selected at the vertex of the curve on the right.

